WRAITH DEFENSE CORPORATION

TECHNICAL DATA: THE M-12 VESTIGE

Long-Range Heavy-Lift Multi-Role UAV System

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Security Level: Commercial/Defense Grade (ITAR/EAR Compliant)

1. MISSION OVERVIEW

The **Wraith M-12 Vestige** is a premier industrial-grade UAV designed to bridge the gap between tactical reconnaissance and heavy logistics. With a class-leading **55kg payload capacity** and an operational **80km transmission range**, the M-12 is the definitive solution for high-stakes of commercial agriculture, search and rescue, and military supply chain support.

2. KEY CAPABILITIES

- **Heavy Payload Integration:** Optimized for 55kg loads, allowing for advanced LiDAR, high-capacity liquid distribution (Agriculture), or medical supply delivery (SAR).
- **Ultra-Long-Range Link:** Powered by the **Wraith S-Link 80**, providing crystal-clear 4K video transmission and telemetry at distances up to 80km.
- **High-Speed Intercept:** Capable speeds exceeding 120km/h for rapid response and time-sensitive deployments.
- **Optic Sovereignty:** Equipped with a 4K Ultra-HD 60FPS camera with 30x Optical Zoom and thermal overlay.
- **Omni-Terrain Deployment:** IP55-rated weatherproofing for operational reliability in extreme environments, including heavy rain, high-salinity coastal zones, and dust-heavy desert theaters.
- **Encrypted Data Sovereignty:** Integrated AES-256-bit hardware encryption, ensuring telemetry and 4K video feeds remain unhackable and secure from electronic intercept.
- **Dual-Battery Redundancy:** Features a "hot-swap" dual-cell power architecture that allows for continuous operation and emergency fail-safe landing in the event of a single-cell malfunction.
- **AI-Driven Obstacle Avoidance:** 360-degree LiDAR and ultrasonic sensor array for autonomous navigation through dense urban corridors and complex forest canopies.
- **Precision RTK Positioning:** Centimeter-level accuracy powered by Real-Time Kinematics (RTK), essential for high-precision mapping and automated agricultural spraying.
- **Modular Payload Interface:** A universal "Quick-Release" rail system allowing operators to switch between 4K gimbals, cargo hooks, or specialized sensors in under 60 seconds.
- **Fiber-Link™ Jamming Immunity:** Optional tethered mode for high-interference zones, providing physical data transmission that is 100% immune to Radio Frequency (RF) jamming.

3. TECHNICAL SPECIFICATIONS

Feature	Specification
Model / Type	M-12 Vestige
Airframe Material	Toray T700 Carbon Fiber + Industrial Fiberglass
Maximum Take-Off Weight (MTOW)	200 kg
Standard Payload Capacity	55 kg (with 23L Fuel Load)
Fuselage Weight	100.5 kg (Airframe Only)
Wingspan / Length	6660 mm / 3856 mm

Power Source	Hybrid Fuel-Electric (8x 6000mAh Battery Array)
Data Link Frequency	1.4 GHz – 1.7 GHz (AES-128 Encrypted)
Operational Range	80 km (Line of Sight)
Ingress Protection	IP54 (Light Rain & Dust Resistant)

3.1 FLIGHT PERFORMANCE ENVELOPE

- **Cruise Speed:** 35 m/s (126 km/h)
- **Maximum Flight Speed:** 42 m/s (151 km/h)
- **Service Ceiling:** 5000 m (MSL)
- **Wind Resistance:** Class 7 (Fixed-wing mode)
- **Thermal Tolerance:** -20°C to +60°C
- **Climb / Descent Rate:** 5 m/s (Ascent) / 3 m/s (Descent)

3.2 MISSION ENDURANCE DAT 4.1 Wraith Neural-Link™ Autonomous Pilotag A

- **Zero Payload:** 420 Minutes (7 Hours)
- **10kg Tactical Load:** 380 Minutes
- **25kg Industrial Load:** 320 Minutes
- **55kg Max Capacity:** 240 Minutes (4 Hours)

4. PROPRIETARY TECHNOLOGIES

4.1 Wraith Neural-Link™ Autonomous Pilotage

Integrated directly into the M-12 onboard flight computer, **Neural-Link™** utilizes a localized AI processor to provide real-time environment mapping. By analyzing high-speed sensor data, the system automatically calculates the safest flight path around obstacles such as power lines, telecommunication towers, and moving aerial hazards. This "Smart-Flight"

logic allows the operator to focus on the mission objective while the drone manages the complexities of low-altitude navigation autonomously.

4.2 Fiber-Link™ Ready

The M-12 is the first in its class to offer a modular Fiber-Link™ port, allowing for physical fiber-optic connection in high-jamming environments where RF signals are compromised.

5. APPLICATIONS

- **Defense:** "Last-mile" logistics, ammunition resupply, and persistent border surveillance.
- **Agriculture:** Precision spraying and multispectral crop analysis over massive acreage.
- **Energy:** Long-range power line and pipeline inspection with 4K high-zoom detail.
- **Public Safety:** Rapid delivery of life-saving equipment (AEDs, rafts) in non-permissive terrains.

6. SYSTEM PACKAGING

Each **M-12 Vestige** unit ships with:

1. **M-12 Airframe** (Carbon Fiber)
2. **Wraith GCS-2** (Dual-Screen Ground Control Station)
3. **Smart-Battery Hub** (High-Density Li-Po Units)
4. **Tactical Transport Case** (Waterproof/Crushproof)

7. OPERATIONAL SUPPORT & COMPLIANCE

- **Defense:** "Last-mile" logistics, ammunition resupply, and persistent border surveillance.

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